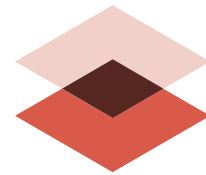


SpieChartGenerator

ERMANS Thomas & Van Hooren Simon



bisa
perspective.brussels 
brussels instituut voor statistiek en analyse



perspective
.brussels 

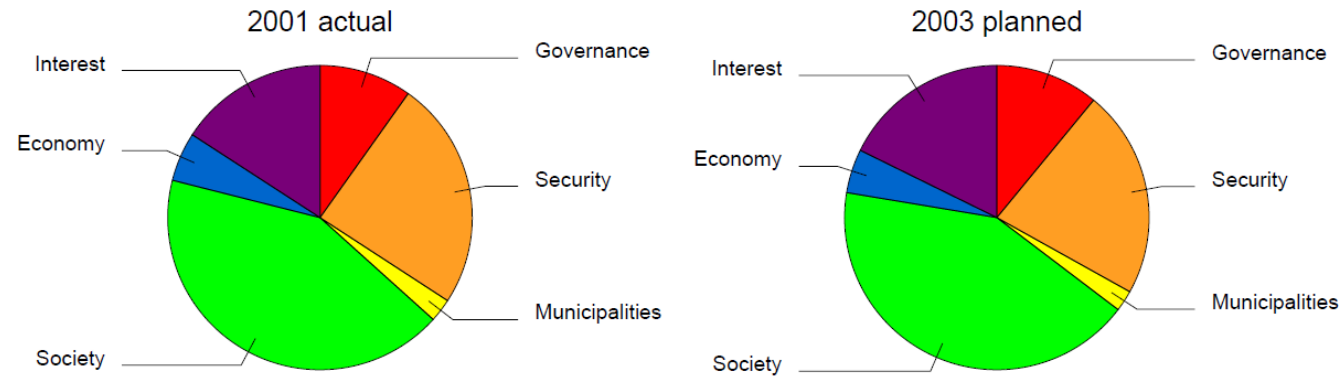
01

WHY DO YOU NEED SPIE CHARTS?

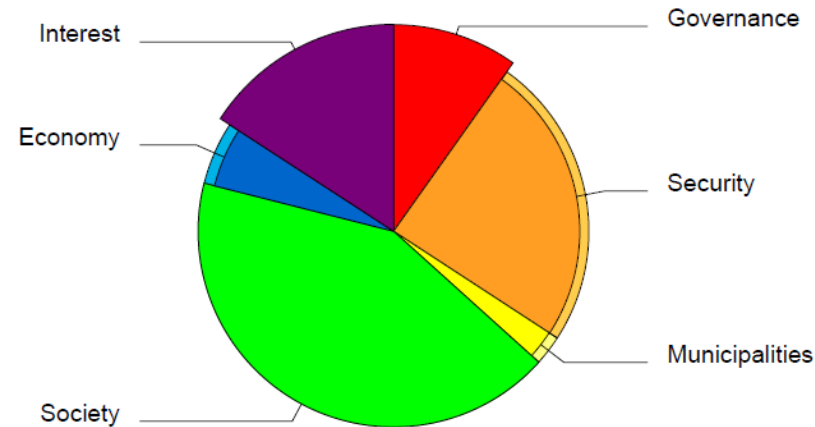


SPIE CHARTS: WHAT ARE THEY?

Pie charts



Spie chart



2001 → simple pie chart

2003 → Spie chart

- Angles are constant but from the 2001 pie chart
- Radii vary to maintain surface w.r.t. simple pie chart

HOW TO CALCULATE THE VARYING RADIUS?

2 distributions, B and C, with $1, 2, \dots, i, \dots, k$ categories

$b_1, b_2, \dots, b_i, \dots, b_k \rightarrow$ observed frequencies for B

$c_1, c_2, \dots, c_i, \dots, c_k \rightarrow$ observed frequencies for C

For a **pie chart** : $Area_{cj} = \frac{\alpha_{cj}}{360} \pi r_c^2$

avec $r_c = 1$

$$\text{et } \frac{\alpha_{cj}}{360} = \frac{c_j}{\sum_{i=1}^k c_i}$$

For a **Spie chart** : $Area_{cj} = \frac{\alpha_{bj}}{360} \pi r_{cj}^2$

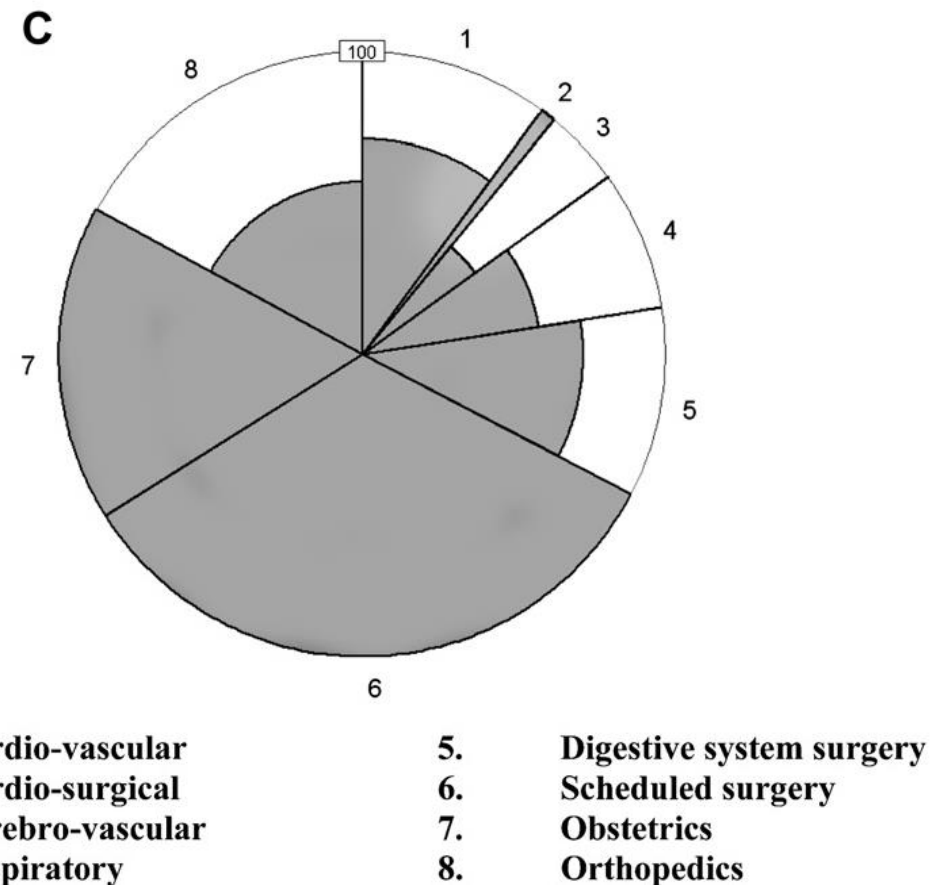
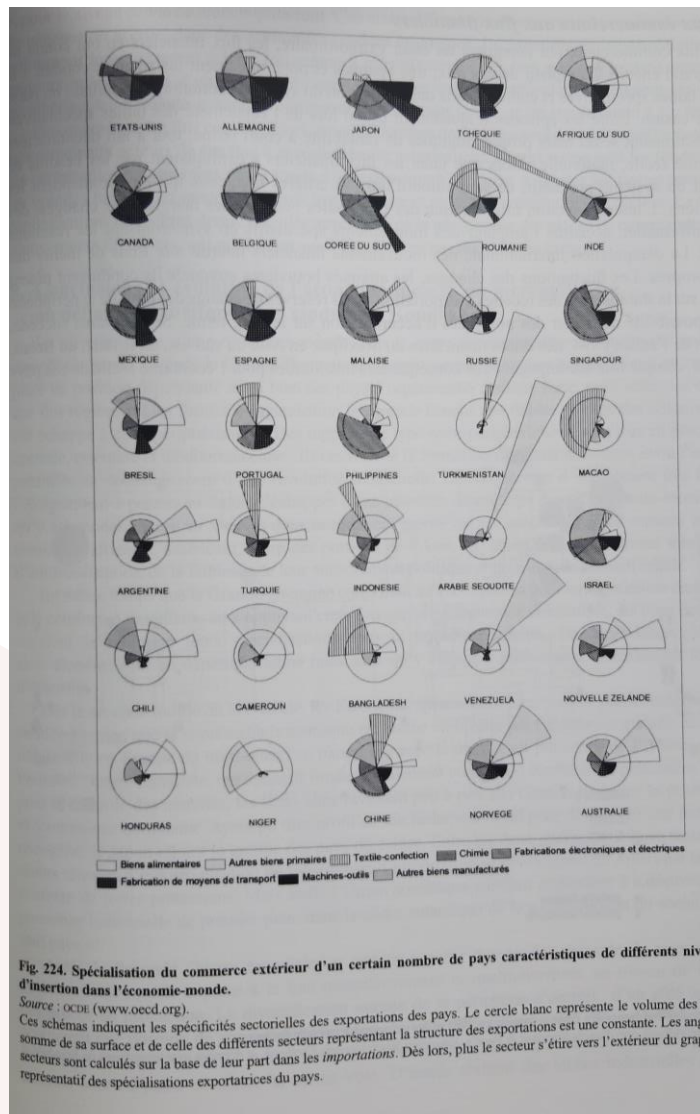
$$\text{Avec } \frac{\alpha_{bj}}{360} = \frac{b_j}{\sum_{i=1}^k b_i}$$

$$\frac{\alpha_{bj}}{360} \pi r_{cj}^2 = \frac{\alpha_{cj}}{360} \pi r_c^2$$

$$\frac{b_j}{\sum_{i=1}^k b_i} r_{cj}^2 = \frac{c_j}{\sum_{i=1}^k c_i}$$

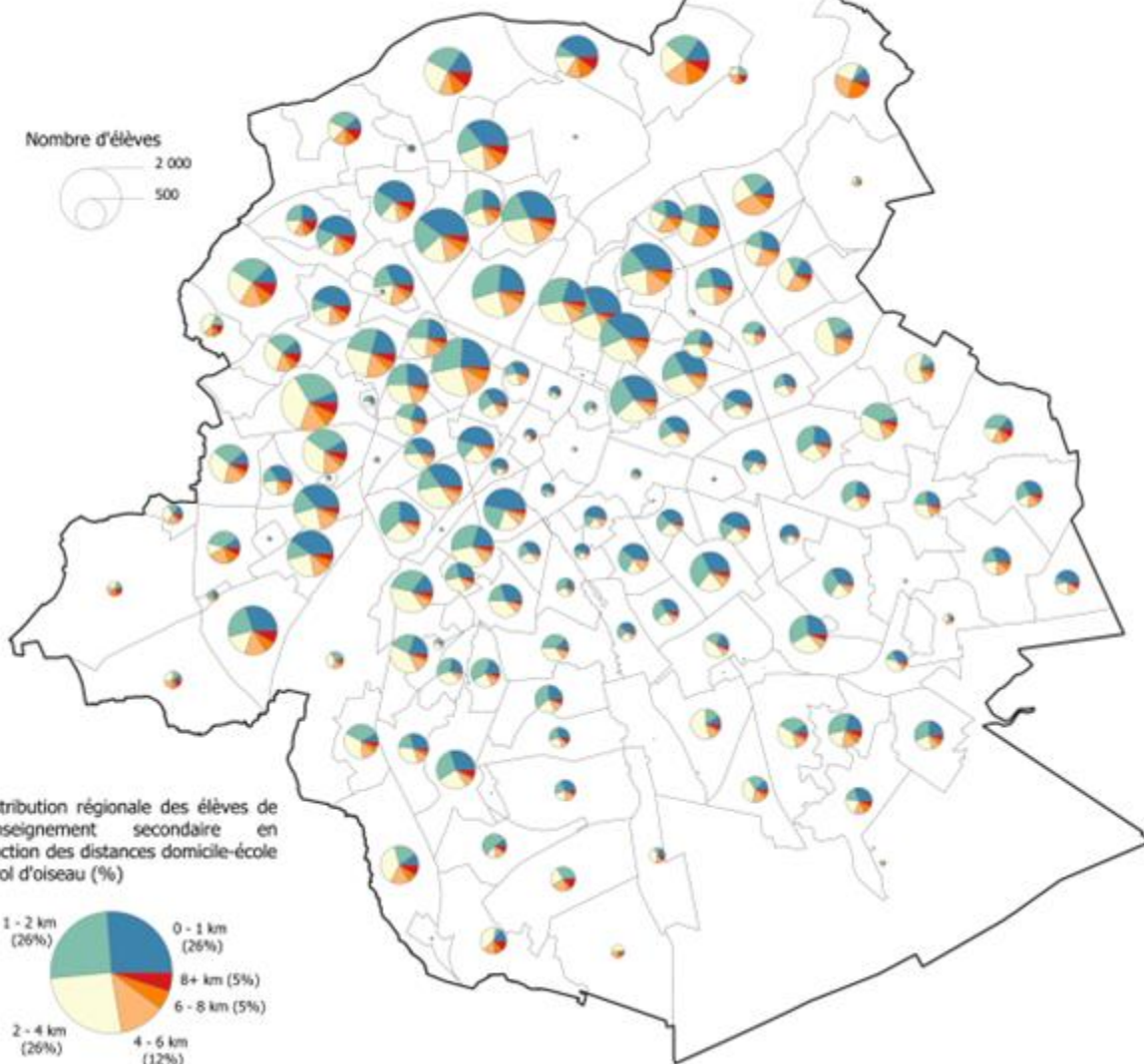
$$r_{cj} = \sqrt{\frac{c_j / \sum_{i=1}^k c_i}{b_j / \sum_{i=1}^k b_i}}$$

OTHER EXAMPLES OF SPIE CHARTS

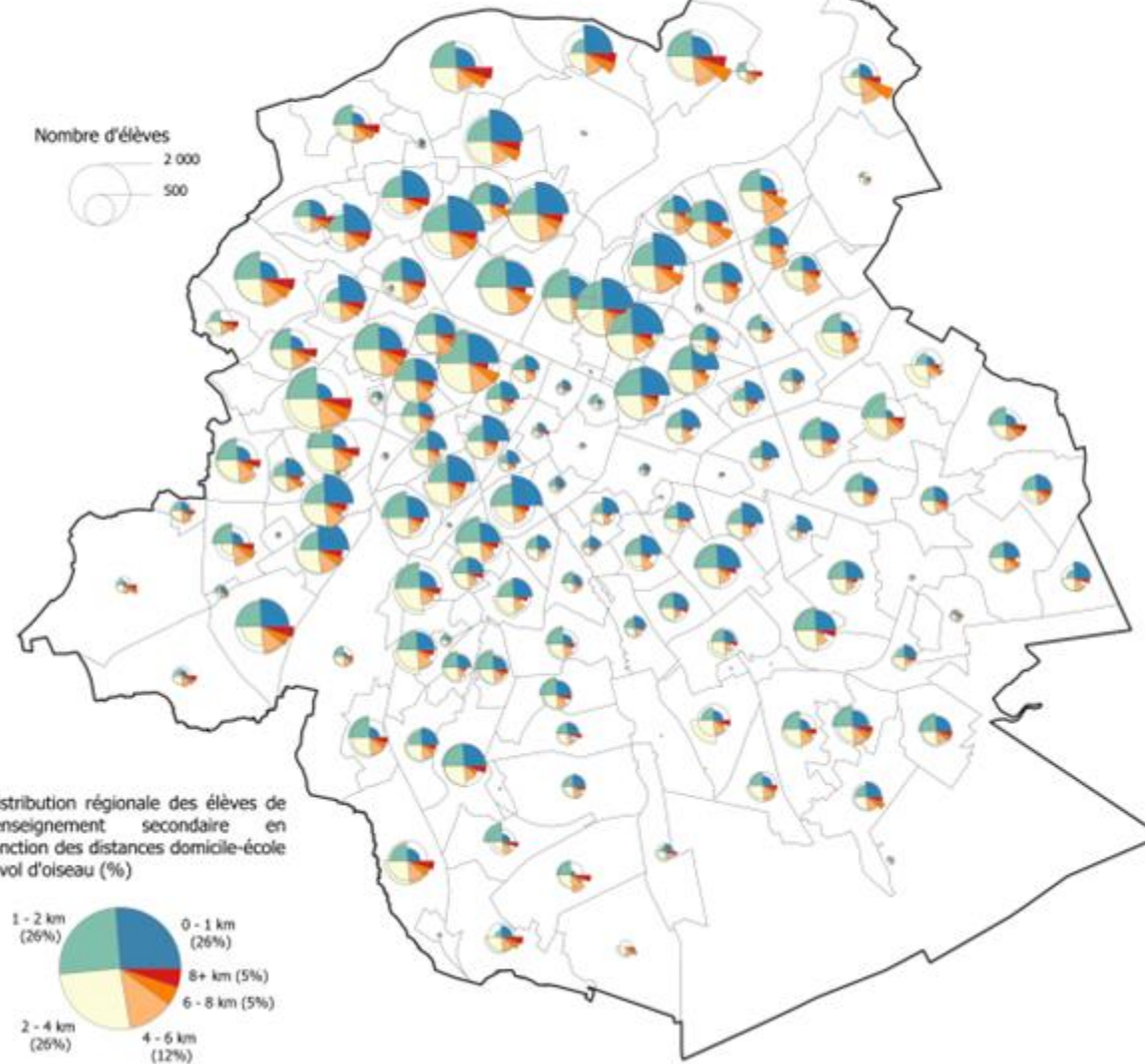


SPIE CHARTS FOR CARTOGRAPHIC VISUALISATION

Diagrammes en camembert (pie charts) - Illustration par les distances domicile-école des élèves scolarisés dans l'enseignement secondaire (ordinaire) par quartier

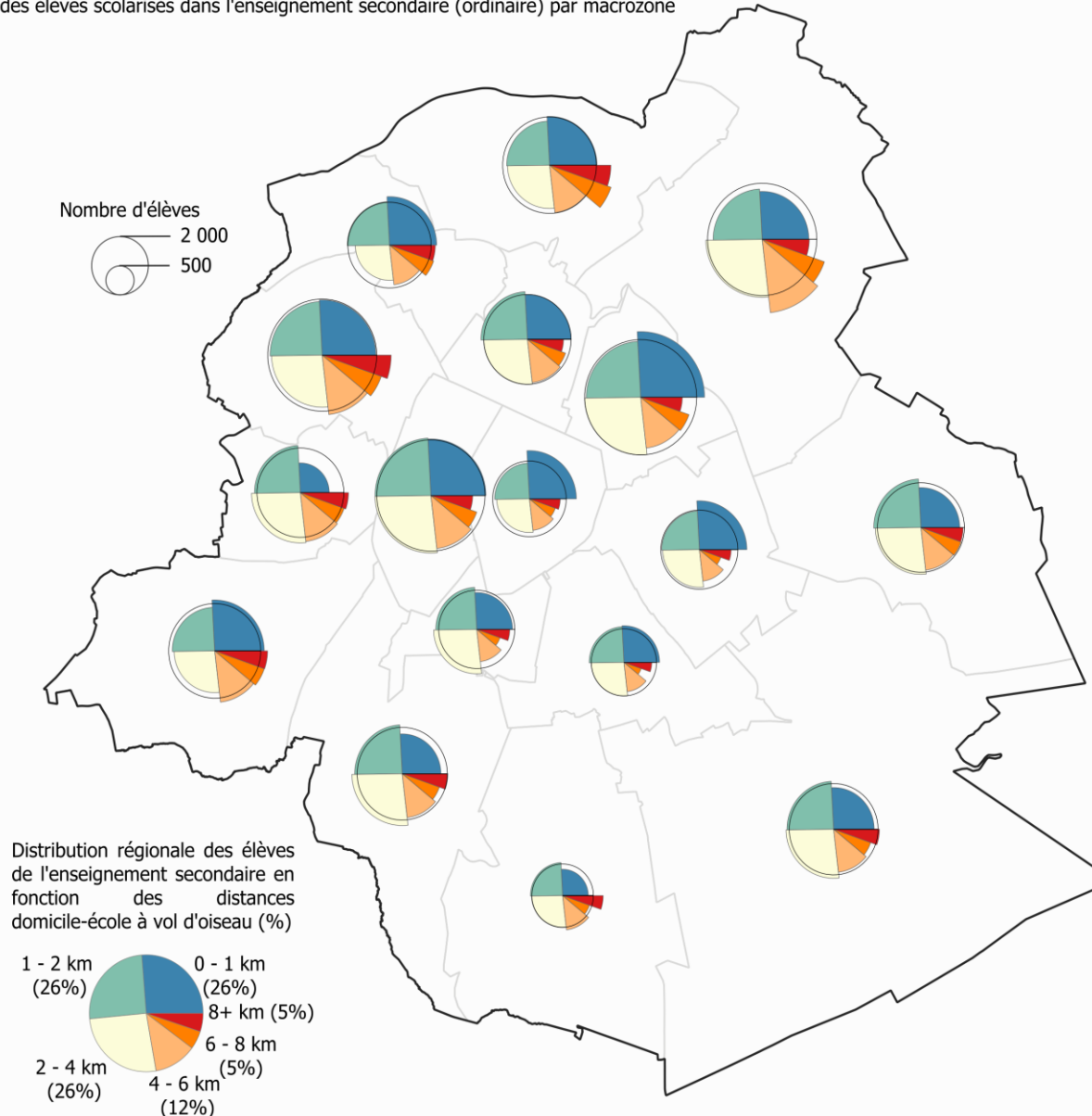


Diagrammes de spécificité (Spie charts) - Illustration par les distances domicile-école des élèves scolarisés dans l'enseignement secondaire (ordinaire) par quartier



SPIE CHARTS FOR CARTOGRAPHIC VISUALISATION

Diagrammes de spécificité (Spie charts) - Illustration par les distances domicile-école des élèves scolarisés dans l'enseignement secondaire (ordinaire) par macrozone



02

WHY A QGIS PLUGIN?



WHY A QGIS PLUGIN?

- Not possible at the moment on QGIS (?)
- Colleagues know excel, less coding
- Learn the matrix and what is happening in QGIS
- Fun to do
- Possible to share the knowledge with other



spie charts in qgis

Tous Images Vidéos Actualités Carte

Protégé Belgique (fr) Filtre parental : modéré Date indifférente

Les résultats pour [pie charts in qgis](#) sont également affichés
Rechercher uniquement pour "spie" charts in qgis ?

YouTube
<https://www.youtube.com/watch?v=IdJuN5JWpU>

How to prepare a Pie Chart/ Diagram in QGIS - YouTube
This video tells how to prepare a Pie Chart/ Diagram in QGIS. A pie chart (or a circle chart) is a circular statistical graphic, which is divided into slices to illustrate numerical...

Geographic Information Systems Stack Exchange
<https://gis.stackexchange.com/questions/439059/qgis-how-to-make-pie-charts-that-depict-...>

QGIS : How to make pie charts that depict quantity on certain ...
Does anyone know how to create a map such as this one in QGIS? In short, if we have a table with list of sites with coordinates and one column with numbers dedicated to µg /g according to which our pie chart is created as depicted on the picture?

GRI
<https://www.gri.msstate.edu/publications/docs/2025/02/16976ChartsQGIS.pdf>

PDF ENHANCING MAPS WITH CHARTS IN QGIS - gri.msstate.edu
In the Expression field, compute the x coordinate of the centroid for each county, then add to it 5000 map units (this will allow to shift the chart to the right):

Geographic Information Systems Stack Exchange
<https://gis.stackexchange.com/questions/15822/performing-statistics-or-simple-pie-chart-in-...>

Performing statistics or simple pie chart in QGIS?
Does anyone know how to perform pie charts based on table values within QGIS and include them in the print outs?

YouTube
<https://www.youtube.com/watch?v=f1tahbxfHGM>

Visualizing Geospatial Data with Pie Charts using QGIS 3
In this tutorial, you will learn how to use pie charts option in QGIS 3 to visualize geospatial data.

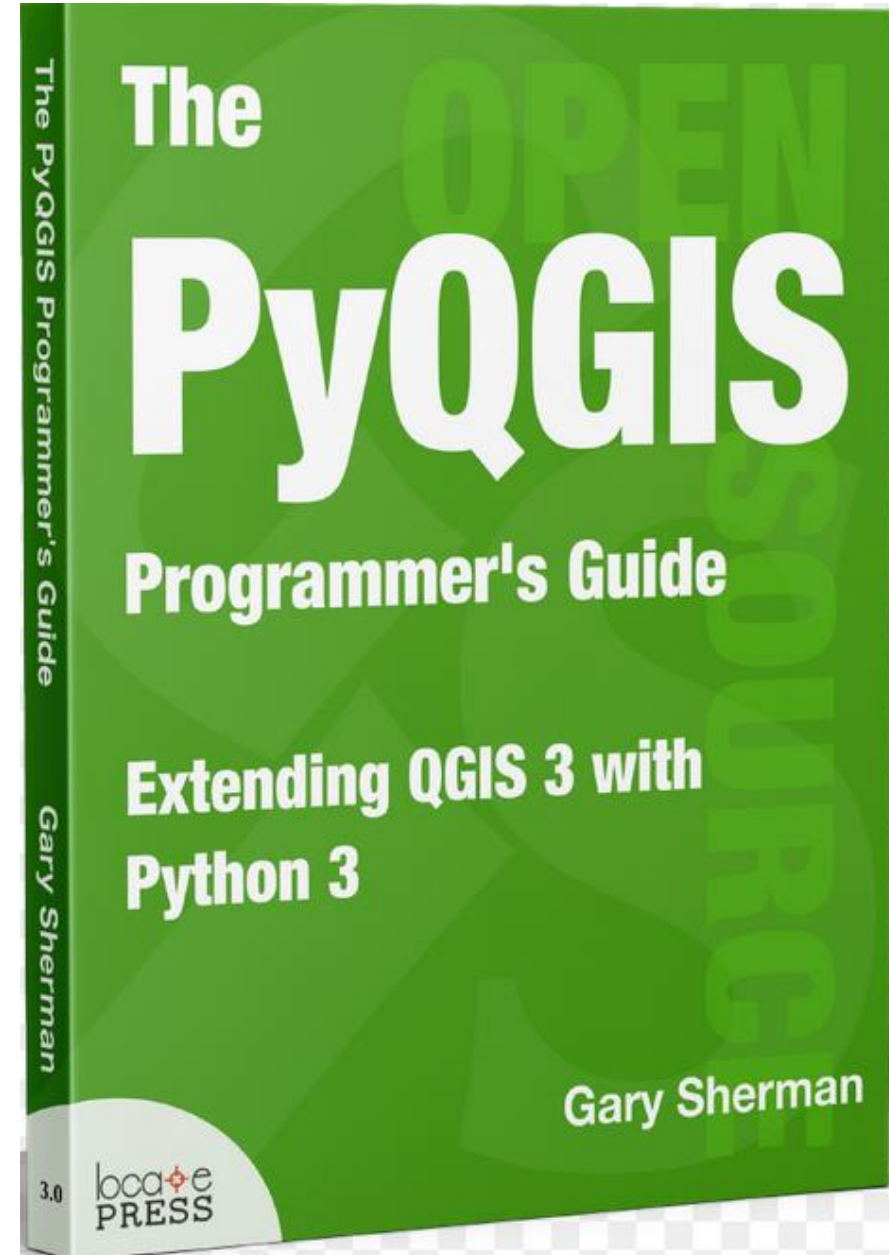
03

HOW TO CREATE A PLUGIN?

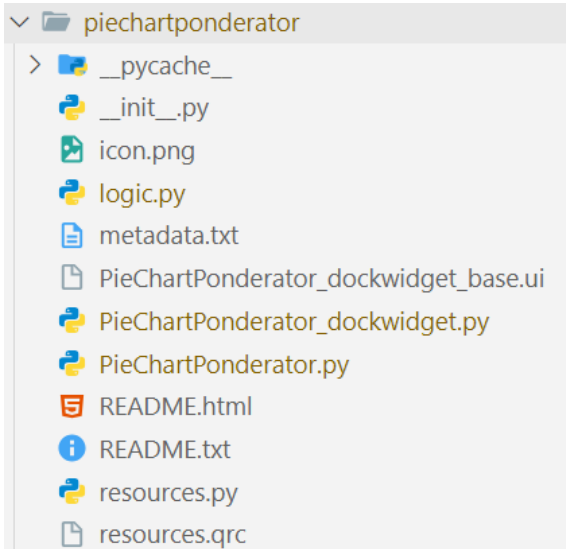


HOW TO CREATE PLUGIN?

- Learn python3 (lots of ressources online - FREE)
- Install the extensions « Plugin Builder » and « Plugin Reloader »
- Use Qt Designer (Optional)
- Use Git for you own sanity
- Prepare your project with some task, and go step by step
- If you know Python and PyQGIS, AI will help you, with some limits



HOW TO CREATE PLUGIN?



```

30 class PieChartLayerBuilder:
31     def __init__(...
57
58     def read_file(self):...
76
77     def expand_categories(self, df):...
106
107     def extract_lon_lat(self, geom_cr
123
124     def compute_angles(self):...
140
141     def get_crs_unit_scale(self, crs:
147
148     def compute_radius(self, group, s
163
164     def create_memory_layer(self):...
170
171     def add_fields(self):...
189
190     def compute_legend_values(self):...
211
212     ...def create_arc(self, center_x, ce
222
223     def build_layer(self):...
371

```

Browse button

```

self.browseButton = QPushButton("Browse")
self.browseButton.clicked.connect(self.browse_csv_file)
self.layout.addWidget(self.browseButton)

```

```

def compute_angles(self):
    # compute global totals from expanded df
    global_angle = (
        self.df.groupby("category")["value"].sum().reset_index(name="total_cat")
    )
    global_total = global_angle["total_cat"].sum()
    if global_total == 0:
        raise ValueError("Total value is zero - cannot compute angles.")

    global_angle["total_cat_pct"] = global_angle["total_cat"] / global_total
    global_angle["start"] = (
        global_angle["total_cat_pct"].cumsum().shift(fill_value=0) * 360
    )
    global_angle["end"] = global_angle["total_cat_pct"].cumsum() * 360

    self.global_angle = global_angle

```

```

def compute_radius(self, group, scale=0.005):
    tot_id = group["value"].sum()
    if tot_id == 0:
        group["radius_scaled"] = 0
        return group
    group["local_cat"] = group["value"]
    group["local_cat_pct"] = group["value"] / tot_id
    # Use global total_cat_pct merged earlier
    group["radius_unscaled"] = np.sqrt(
        group["local_cat_pct"] / group["total_cat_pct"] * tot_id
    )
    group["radius_scaled"] = (
        np.sqrt(group["local_cat_pct"] / group["total_cat_pct"] * tot_id) * scale
    )
    return group

```

04

USAGE



USAGE

Inputs:

- CSV file (geom column and categories)

	A	B	C	D	E	F	G	H	I	J	K
1	quartier_id	geom	quartier_code	quartier_nom_fr	nb_eleve	distvo_0_1_nb	distvo_1_2_nb	distvo_2_4_nb	distvo_4_6_nb	distvo_6_8_nb	distvo_8_plus_nb
2	1	POINT (148867.2128842451 170703.6231507622)	1	GRAND PLACE	179	114	23	26	9	7	0
3	2	POINT (148322.2601882884 171178.197999726)	2	DANSAERT	712	492	129	44	21	14	12
4	3	POINT (148676.7062940269 171611.1487242051)	3	BEGUINAGE - DIXMUDE	570	339	124	51	30	15	11
5	4	POINT (149229.3012354871 171334.9566466708)	4	MARTYRS	178	91	46	17	15	3	6
6	5	POINT (149753.1697806571 171109.902372572)	5	NOTRE-DAME AUX NEIGES	217	115	58	28	10	2	4

- EPSG (impact the calculations)
- Scale

USAGE

Output:

- The spie charts with a color per category
- A surimposed circle for better understanding (still needs improvement)
- The values to manually create ellipses for the legend

☒ ☐ Total_Circles_secondaire 15

▼ ☒ ☐ seconaire 15

☒ ☐ distvo_0_1_nb

☒ ☐ distvo_1_2_nb

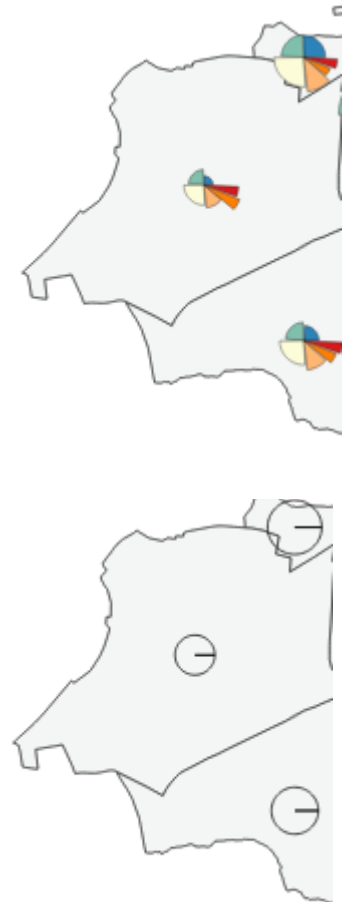
☒ ☐ distvo_2_4_nb

☒ ☐ distvo_4_6_nb

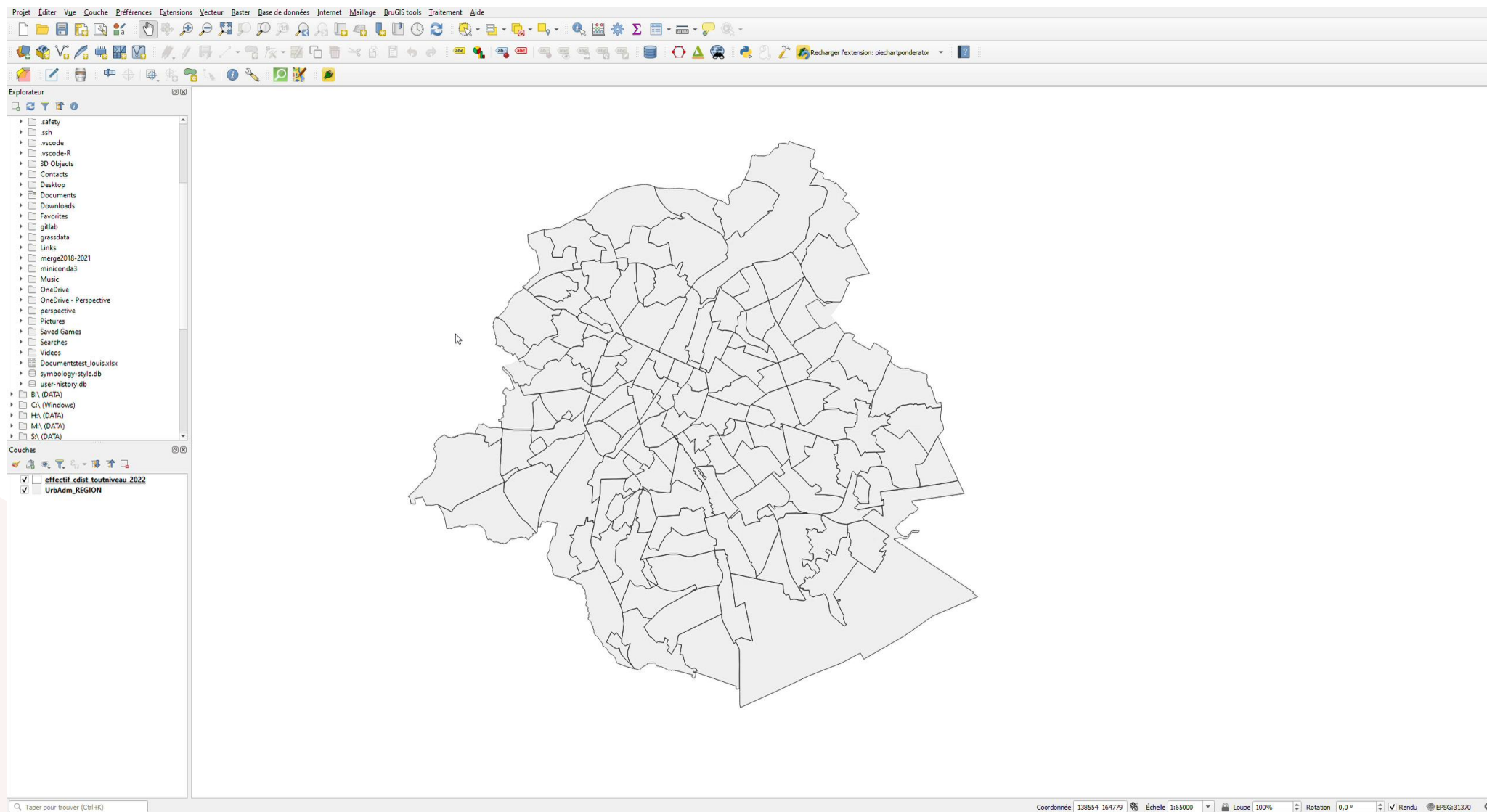
☒ ☐ distvo_6_8_nb

☒ ☐ distvo_8_plus_nb

	lue of the categori	Millimeter
Smallest	0.0	0.00
Median	97.0	13.00
Largest	2408.0	16.72



USAGE



05

FUTURE



FUTURE

- **Better legend management**
- **Calculate all values, for filter, to improve visualization**
- **Choice of EPSG from anywhere**
- **Public code**
- **Other pie charts plugin**